Research on Multimodal Transportation Path Optimization Based on Improved Genetic Algorithm

2023 7th International Conference on Transportation Information and Safety (ICTIS) | 979-8-3503-0853-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICTIS60134.2023.10243715

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***Abstract*—For the multi-objective multimodal transportation path optimization problem, based on the analysis of the objective function, the multi-objective path optimization model of container multimodal transportation with class period limitation is constructed by considering the four factors of transportation time, transportation cost, carbon emission, and multimodal logistics service quality, taking into account the different preferences and differentiated demands of different multimodal transportation participating decision makers, and setting different objective functions. The model is solved by a modified genetic algorithm based on the TOPSIS method, and the optimal transportation scheme under different weight combinations is obtained by solving it with MATLAB software using the Chengdu-Shanghai transportation network as an example.**

Keywords—Multimodal Transportation; Path optimization; TOPSIS Method; Improved genetic algorithm

I. INTRODUCTION

Logistics plays a crucial part in the enhancement of social and economic advantages as a result of the social economy's ongoing development and the accelerated development of e-commerce and the Internet. Multimodal transportation may not only fully utilize the benefits of multiple modes of transportation, but also strengthen their interoperability in order to accomplish the goals of lowering costs, shortening transit times, and raising the caliber of logistics services. As a result, multimodal transportation is taking on more significance in the world of transportation, and research on the subject of route optimization for multimodal transportation is booming.

According to the findings of numerous academic studies on multimodal transportation, the primary issue is the optimization of multimodal transportation routes. The majority of studies evaluate and investigate the multimodal path optimization problem by using pertinent theoretical understanding, building corresponding mathematical optimization models, and developing solution algorithms in the context of various influencing elements.Mouna Mnif et al. [1] constructed a multi-objective optimization model with transportation cost minimization and transportation time minimization as the optimization objectives, which was solved using a multi-objective fireworks algorithm, from which the final results of the Pareto solution set were obtained. By comparing the results of the analysis, it was concluded that the heuristic algorithm is more efficient in solving the problem. Luo et al. [2] constructed a multi-objective optimization model based on the analysis of multimodal transportation costs and time costs and solved the optimal path using the Floyd-GA hybrid algorithm. Song [3] constructed a low-carbon multimodal transportation path optimization model with the objectives of carbon emission cost of cargo transportation, transportation cost, and transshipment cost under the time constraint, considering carbon emission and transportation cost, and obtained the optimal multimodal transportation path scheme by using a genetic algorithm to solve. Xie et al. [4] construct a 0－1 integer planning model with a time window constraint for whole-vehicle logistics and multimodal transportation with the objective of minimizing logistics cost, taking transportation time as the constraint, and considering the storage cost due to early arrival and the penalty cost due to delayed arrival. Lv et al. [5] constructed a hybrid time window by considering a combination of soft and hard time windows, used this as a restriction to construct a path selection model with the objective of minimizing the total cost, and analyzed its solution using an ant colony algorithm on this basis. Wan et al. [6] constructed a multi-objective path optimization model with the minimization of all three as the optimization objective by considering transportation time, carbon emission, and transportation cost and solved it by using the improved fireworks algorithm. The results obtained were good enough to verify the effectiveness of the model algorithm. Jiang et al. [7] constructed a path optimization model with overall consideration of transportation time, carbon emission, and transportation cost based on the study of the influencing factors of carbon emission and analyzed the influence of different decision contexts on the decision of the optimal path. Liu et al. [8] construct a model with the objective of minimizing the total cost under the condition of the existence of carbon emission constraints and design algorithms to solve the model. Considering that the customer's demand is not deterministic, Deng et al. [9] developed a model with the objective of minimizing carbon emissions as well as the total cost and used a combination of simulated annealing as well as particle swarm algorithms to solve the problem.

In summary, from the scholars' research results on multimodal transportation path optimization problems, at present, most of the research on container multimodal transportation problems is based on time, cost, and other perspectives to construct transportation models. However,

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some past research only considered the time consumption of changing transportation modes for cargo, which is not close to reality. It is very important and relevant to take this limitation into account since, in general, the time spent waiting for the appropriate shift is much more expensive than the time spent changing the mode of transportation. In addition, in today's highly developed logistics technology, the quality of logistics service should be the new thinking direction, and then consider multi-objective optimization problem research with cost, time, and logistics service quality. Therefore, in the context of green transportation, this paper considers the construction of a multimodal transportation path optimization model with the objectives of transportation cost, time, carbon emission, and logistics service quality and designs algorithms for solving them to provide references for decision-makers of the scheme.

II. MULTIMODAL TRANSPORTATION PATH OPTIMIZATION MODEL CONSTRUCTION

A. Problem description

Assume that a batch of goods is to be transported from the starting city O to the end city D. From the starting city O, you can choose any one of the three modes of transport between road, rail, and waterway; in the middle, you can choose n cities for the transit hub node, thus constructing a complete path from the starting point to the end. In any connection between the two cities nodes, road, rail, and water transport are randomly arranged to be chosen. Excluding the origin and destination points, each node can realize the switching of goods between different transportation modes; in addition, there is a time demand for the arrival of goods by customers during transportation.

B. Conditional assumptions

1) At most, only one transit can occur at a transit node, and only from one mode to another.

2) When cargo is transported, the mode of transportation does not change suddenly, and if the mode of transportation is changed, the change operation is carried out after arrival at the node city.

3) Only the cargo is considered to be transported in uniformed boxes, and no distinction is made between the types of cargo.

4) When at the node, if its next mode of transport is railroad or waterway transport, it needs to wait until the departure moment of the corresponding train or schedule, and if its next mode of transport is road transport, it does not need to wait.

5) The cargo is considered as a whole during transportation as well as transit, i.e., the quantity of the cargo remains the same.

6) During the transportation process, the transport speed of all modes of transport always maintains a constant value and does not change.

7) The transportation capacity of all modes of transportation in each node city can meet the actual transportation needs..

C. Determination of objective functions

1) Analysis of time function

The time function in this paper consists of three parts: the transport time of the container, the time spent by the cargo for transit operations, and the time needed to wait for the departure of trains and ships.

a) Transportation time

The transportation time between hub node cities is determined by the distance and speed between the nodes. The calculation formula is as follows:

$$t_{ij}^{k} = \frac{d_{ij}^{k}}{v_{k}} \tag{1}$$

In equation (1):t_{ij}^{k} denotes the transportation time that needs to be spent if the container cargo is chosen to be transported from node city i to node city j through k transportation modes;d_{ij}^{k} denotes the transport distance that needs to be passed if the container cargo is chosen to be transported from node city i to node city j via k transport mode;v_k denotes the speed of transportation process through k transportation mode.

The total transportation time T_1 is calculated by the following equation:

$$T_1 = \sum_{i,j\in N}\sum_{k\in M} t_{ij}^{k}\cdot x_{ij}^{k} \tag{2}$$

In equation (2):N denotes the set of all urban nodes;M denotes the set of all modes of transportation;x_{ij}^{k} is the decision variable, which indicates that it takes the value of 1 when the event cargo is transported from node i to node j by transportation mode k, and 0 otherwise.

b) Transit time

Transit time mainly refers to the time required for the changeover operation between the modes of transportation if the changeover is to occur at the node, and the formula for calculating the transit time T_2 is as follows:

$$T_2 = \sum_{i\in N}\sum_{k,l\in M} Q\cdot t_i^{kl}\cdot y_i^{kl} \tag{3}$$

In equation (3):Q denotes the number of containers;t_i^{kl} denotes the unit time taken to convert from transportation mode k to l at node city i;y_i^{kl} is the decision variable, which indicates that when the cargo is in node city i, it takes the value of 1 if the event transport mode k converts to l, otherwise it takes the value of 0.

c) Waiting time

The waiting time W_{ij}^{l} is calculated as follows:

$$W_{ij}^{l} = \sum_{i\in N}\sum_{k,l\in M}\left(L_i^{l} - T_i^{k} - Q\cdot t_i^{kl}\cdot y_i^{kl}\right) \tag{4}$$

It can also be expressed as follows:

$$W_{ij}^{l} = \begin{cases} ep_{ij}^{l} - T_i^{k} - Q\cdot t_i^{kl}\cdot y_i^{kl}, & l = 2,3 \\ 0, & l = 1 \end{cases} \tag{5}$$

In equations (4) and (5):L_i^{l} denotes the moment when the cargo container leaves node i by transport mode l;T_i^{k} denotes

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the moment when the cargo container enters node i by transport mode k;W_{ij}^{l} denotes the moment when the cargo waits at node i by transport mode l;ep_{ij}^{k} denotes the moment of departure of cargo container from node i to node j in the pth shift by transportation mode k (rail or waterway).

2) Analysis of cost function

In the model considered for construction in this paper, the costs incurred in multimodal transportation consist of the following components: transportation costs between nodes, costs arising from transit occurring in node cities, waiting costs caused by early arrival, and penalty costs resulting from not arriving at the destination within the time frame specified by the customer.

a) Transportation cost

The formula for calculating transportation cost C_1 is as follows:

$$C_1 = \sum_{i,j\in N}\sum_{k\in M} Q\cdot c_{ij}^{k}\cdot d_{ij}^{k}\cdot x_{ij}^{k} \tag{6}$$

In equation (6):c_{ij}^{k} denotes the unit transportation cost for delivery from node i to node j by transportation mode k.

b) Transit cost

The formula for calculating the transit cost C_2 is as follows:

$$C_2 = \sum_{i\in N}\sum_{k,l\in M} Q\cdot c_{i}^{kl}\cdot y_{i}^{kl} \tag{7}$$

In equation (7):c_i^{kl} denotes the unit transit cost of converting the transport mode from k to l at city node i.

c) Waiting cost

During the waiting time, goods stacked at the site will incur certain stacking costs. The waiting cost C_3 is calculated as follows:

$$C_3 = \sum_{i,j\in N}\sum_{k\in M} Q\cdot c^{k}\cdot W_{ij}^{k} \tag{8}$$

In equation (8):c^k denotes the unit waiting cost of goods waiting by transportation mode k.

d) Penalty cost

Penalty costs are incurred if the goods do not arrive successfully at their destination within the time frame specified by the customer. The penalty cost P is calculated by the following formula:

$$P=\begin{cases} a(ET-T)Q, T\le ET \\ 0, ET\le T\le LT \\ b(LT-T)Q, T\ge LT \end{cases} \tag{9}$$

In equation (9):P denotes the penalty cost function;ET denotes the earliest arrival time of the specified goods;LT denotes the specified latest time of arrival of the goods;a denotes the unit waiting cost in the penalty function;b denotes the unit penalty cost in the penalty function.

3) Analysis of carbon emission

The formula for calculating E_1 carbon emission in the transportation process is as follows:

$$E_1 = \sum_{i,j\in N}\sum_{k\in M} Q\cdot e^{k}\cdot d_{ij}^{k}\cdot x_{ij}^{k} \tag{10}$$

The calculation formula of carbon emission in the transit process is as follows:

$$E_2 = \sum_{i\in N}\sum_{k,l\in M} Q\cdot e^{kl}\cdot y_{i}^{kl} \tag{11}$$

In equations (10) and (11):e^k denotes the unit carbon emission of cargo container transported by transport mode k; e^{kl} denotes the unit carbon emission of converting from transport mode k to transport mode l.

4) Analysis of logistics service quality function

This paper considers using customer satisfaction to quantify the quality of logistics service from the following two aspects of analysis:

a) Timeliness

In this paper, the time window is fuzzified, and the fuzzy time window is represented by a trapezoidal fuzzy number. Set the interval [ET,LT] as the time range specified by the customer; only within this time range can the goods arrive at the destination to satisfy the customer and not make the customer's satisfaction decrease, and its value is 1; set the interval [Tmin,Tmax] as the time range that the customer can accept; that is, if the goods arrive at the destination outside this time range, the customer's satisfaction is the lowest, and its value is 0; And if the goods arrive at the destination within the interval [LT,Tmax-LT] and the interval [Tmin,ET-Tmin], the customer's satisfaction decreases with the increase of the distance between the specific moment of arrival at the destination and the interval [ET,LT]. In summary, the customer's satisfaction HT for timeis expressed as an affiliation function, as shown in equation (12) below.

$$H_T=\begin{cases} 1, \quad T\in(ET,LT) \\ \dfrac{T-Tmin}{ET-Tmin}, T\in(Tmin, \quad ET) \\ \dfrac{Tmax-T}{Tmax-LT}, T\in(LT,Tmax) \\ 0, \quad else \end{cases} \tag{12}$$

b) Safety

The so-called safety, that is, the security of multimodal transport, cargo damage in various modes of transport, and cargo difference reflect the ability of different modes of transport to protect the safety of goods.Security is an important symbol reflecting the quality of transport services. The occurrence of serious cargo damage and cargo differences in the transport of goods, resulting in the failure to safely arrive, will cause customers to suffer huge losses, resulting in customer dissatisfaction and reducing the level of logistics services perceived by customers. According to the relevant requirements of our country, if the cargo loss caused by transportation exceeds the value of 0.5%, then the insurance company should pay compensation for this transportation service. Set the cargo loss rate Sa for the customer's satisfaction of the maximum rate of cargo loss. If

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the rate of cargo loss of transport routes S is lower than Sa, then the customer's satisfaction is 1. A cargo loss rate that is higher than this value will reduce customer satisfaction. Take the claim critical rate of cargo loss value of Smax = 0.5% so that the customer can accept the maximum rate of cargo loss. In summary, the customer satisfaction HS for security is based on the subordinate degree function, as shown in the following equation (13).

$$H_S = \begin{cases} 1, S \in (0, Sa) \\ \dfrac{Smax - S}{Smax - Sa}, \quad S \in (Sa, \ Smax) \\ 0, \quad S \in (Smax, +\infty) \end{cases} \tag{13}$$

In summary, the logistics service quality LSQ expression is as follows:

$$LSQ = H_T + H_S \tag{14}$$

D. Model Building

Through the above analysis, the objective functions of the multi-objective optimization model constructed in this paper are specifically the following four:

$$\min Z_1 = \min(T_1 + T_2 + \sum W_{ij}^{k}) \tag{15}$$

$$\min Z_2 = \min(C_1 + C_2 + C_3 + P) \tag{16}$$

$$\min Z_3 = \min(E_1 + E_2) \tag{17}$$

$$\max Z_4 = \max(LSQ) \tag{18}$$

Equations (15)-(18) indicate the shortest transportation time, the lowest cost, the smallest carbon emission,and the highest quality of logistics service, respectively.

The constraints of the model are as follows:

$$\sum_{k \in M} x_{ij}^k \le 1, \quad \forall i, j \in N \tag{19}$$

$$\sum_{k,l \in M} y_i^{kl} \le 1, \quad \forall i \in N \tag{20}$$

$$\sum_{j \in A(i)} \sum_{k \in M} x_{ij}^k - \sum_{j \in B(i)} \sum_{k \in M} x_{ji}^k = \begin{cases} 1, \\ i \text{ is the starting point} \\ 0, \\ i \text{ is the intermediate node} \\ -1, \\ i \text{ is the end point} \end{cases} \tag{21}$$

$$\sum_{j \in B(i)} x_{ji}^k + \sum_{j \in A(i)} x_{ij}^l \ge 2 y_i^{kl} \tag{22}$$

$$T_i^k = L_{i^-}^k + \frac{d_{ij}^k}{v^k}, \quad \forall i^-, i \in N \tag{23}$$

$$L_i^l = \begin{cases} T_i^k + Q \cdot t_i^{kl} \cdot y_i^{kl}, & l = 1 \\ ep_{ij}^l, & l = 2,3 \end{cases}, \tag{24}$$
$$\forall i, j \in N; \forall k, l \in M; \forall ep \in E$$

$$T_i^k + Q \cdot t_i^{kl} \cdot y_i^{kl} < en_{ij}^l, \tag{25}$$
$$\forall i, j \in N; \forall k, l \in M; \forall en \in E$$

$$ep_{ij}^l = \min(en_{ij}^l), \tag{26}$$
$$\forall i, j \in N; \forall l \in M; \forall ep, en \in E$$

$$W_{ij}^l = \begin{cases} ep_{ij}^l - T_i^k - Q \cdot t_i^{kl} \cdot y_i^{kl}, & l = 2,3 \\ 0, & l = 1 \end{cases}, \tag{27}$$
$$\forall i, j \in N; \forall k, l \in M; \forall ep \in E$$

$$x_{ij}^k, y_{ij}^k = \{0,1\} \tag{28}$$

Among them, constraint (19) indicates that when cargo is transported between node i and node j, its transportation mode can and can only choose one of the three, constraint (20) indicates that at most one transit operation can be performed at node i, constraint (21) ensures the balance between traffic input and output at all nodes, constraint (22) ensures the continuity of the process of multimodal transportation of cargo containers Constraint (23) indicates the time relationship between the moment of arrival of cargo at node i, constraint (24) indicates the time relationship between the moment of departure of cargo from node i, constraints (25) and (26) together determine the time of departure of cargo by rail or waterway at node i, constraint (27) indicates the waiting time due to the constraint of schedule, and Constraint (28) indicates that the value of the restricted decision variable can only be chosen between 0 and 1.

III. Design of Solution Algorithm

A. Encoding and decoding

When a genetic algorithm is used to solve path optimization in multimodal transportation, the choice of its chromosome encoding method is largely determined by the real situation of the problem. Due to the complexity of multimodal transportation networks and the diversity of transportation modes, the traditional method of binary encoding of decision variables in the problem will result in slow convergence of the algorithm due to the large number of variables. In this paper, we consider the hierarchical treatment of a multimodal transportation network and encode and decode it from two aspects: the selection of nodes through which transportation paths pass and the selection of transportation modes between two nodes.

If the multimodal transportation network can be divided into n layers, excluding the layer with only one node per layer, it is assumed that there are m layers, so n-m decisions need to be completed to select the nodes through which the transportation path passes, and there are multiple transportation modes between each two adjacent layers, but the nodes selected by two adjacent layers have been fixed in the previous decisions, so the next only needs to make the selection decision between each two adjacent layers transportation mode between each two adjacent levels, a total of n-1 decisions are made, to which the model is transformed into an optimization model with (n-m+n-1) decision variables.

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The coding and decoding processes are illustrated in Fig. 1 as an example.

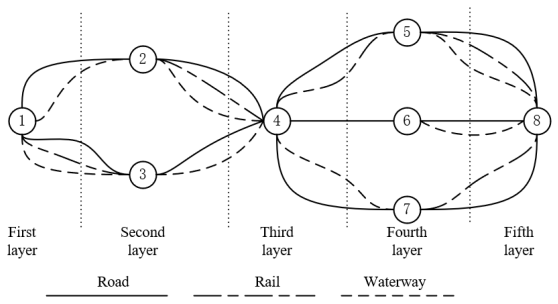


Fig. 1. A multimodal transportation network diagram

First, this multimodal transportation network can be divided into five layers, as shown in Fig. 1, where only one node exists in the first, third, and fifth layers, so two decision variables are used to indicate the selection of nodes through which the transport path passes and four decision variables are used to indicate the selection of transport modes. Then, individuals with a chromosome length of 6 are generated, and the first 2 genes indicate the selection of nodes, and the last 4 genes indicate the selection of transportation modes. Floating-point numbers are used to encode the genes, i.e., a randomly generated four-digit decimal number with values in the range of (0,1), such as 0.3489, etc. The (0,1) interval is equipartitioned according to the number of nodes present in each layer, such as two nodes in the second layer, then two equipartitions, and three nodes in the fourth layer, then three equipartitions. The decoding idea then selects the node corresponding to the interval according to the interval in which the value of the corresponding random decimal at its genetic position is located. Similarly, the selection of the transport mode is the same as above. A random chromosome is generated, and its encoding and decoding are schematically shown in Fig. 2.

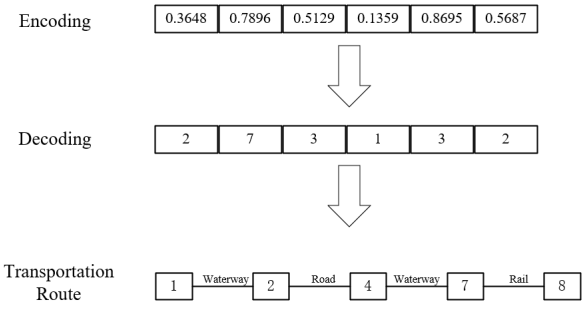


Fig. 2. Schematic diagram of chromosome coding and decoding

B. Adaptability calculation and evaluation based on TOPSIS method

For this paper, the constructed model is a multi-objective planning model, so it is considered to be transformed into a single objective to solve it. In the process of multi-objective optimization of multimodal container transportation, it is considered that different participants in multimodal transportation, such as shippers, operators, managers, etc., have different preferences for different objectives when making route selection decisions.

According to the pertinent literature, multimodal transport managers, in the process of route decision, take more into account the social and environmental impacts of transportation activities [10], so they place a greater emphasis on transportation solutions with lower carbon emissions. For multimodal transport operators, such as carriers, in the process of route decision, they are primarily concerned about their economic benefits, while at the same time, in order to attract customers, they pay attention to transport time, but compared to the cost of transportation, the issue is less important. For multimodal consumers such as shippers, they tend to pay more attention to transportation time and transportation cost in the process of route decision-making and pay the least attention to carbon emissions [11]. In reality, due to the differences in the characteristics of various types of goods, the needs of each goal of multimodal transportation are different. For example, for goods such as fresh products, shippers focus more on transportation time when making decisions, while for general cargoes such as electronic parts, they are more focused on transportation costs. Therefore, when choosing the route, different weights are given to each target factor to solve the multi-objective problem as a single objective. Customers can also determine the corresponding weights according to their own needs and the importance of different target factors in order to solve the problem of differentiated requirements and different emphasis on multimodal transportation solutions for different customers. However, due to the different scales of each objective involved in the paper and the fact that the quality of logistics services in the objective function is measured by satisfaction, the solution cannot be simply solved by a linear weighting method.

The TOPSIS (Technique for Order Preference by Similarity to an Ideal Solution) method is a common and effective method in multi-objective decision analysis, with a simple concept, a clear calculation process, high operability, etc. The TOPSIS method has been widely used in multi-objective decision making and multi-objective optimization, so this paper selects this method for fitness evaluation, and the specific steps are as follows:

1) Determination of the positive ideal solution

The mathematical software Matlab is used to find out the optimal solutions of the four single-objective optimization problems under the constraints of this paper, where logistics service quality is a benefit-type indicator, and the larger its value, the better the solution obtained, while transportation cost, transportation time, and carbon emission are cost-type indicators, and the smaller their values, the better the solution obtained. Therefore, to facilitate the calculation process, the value of logistics service quality is taken as the reciprocal, and in order to prevent errors in the subsequent writing of the program, the denominator is operated by adding 1, as shown in equation (29). The row vector Z+, which is composed of the optimal solutions of the four objectives, is defined as the positive ideal solution of the multi-objective optimization.

$$\max LSQ = \min \frac{1}{LSQ+1} \tag{29}$$

$$Z^{+} = [Z_1 \min \square\ Z_2 \min \square\ Z_3 \min \square\ Z_4 \min] \tag{30}$$

2) Determination of the negative ideal solution

The mathematical software Matlab is used to find the inferiority solutions of the four single-objective optimization

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problems under the constraints of this paper, and the row vector Z^-,consisting of the worst solutions of the four objectives, is defined as the negative ideal solution of the multi-objective optimization.

$$Z^- = [Z_1\max\square\ Z_2\max\square\ Z_3\max\square\ Z_4\max] \quad (31)$$

3) Determination of the range of values of the objective function

The range of values of the objective function for a feasible solution of any single-objective optimization problem i is [Zimin,Zimin], i=1,2,3,4.

4) Normalization process

Due to the non-uniformity of the magnitudes of the multiple objectives considered in this paper, their target values have different degrees of influence on the multi-objective optimization, so in order to eliminate the influence of multiple target values of different magnitudes, they need to be normalized to unify the magnitudes, and the calculation formula for normalization is as follows:

$$Z_i' = \frac{Z_i^{\max} - Z_i}{Z_i^{\max} - Z_i^{\min}} \quad (32)$$

5) Calculate the distance between the feasible solution and the positive and negative ideal solution

The core of the Approximating Ideal Point method is to determine the superiority of the solution by calculating the distance between each feasible solution and the positive and negative ideal solutions; the closer to the positive ideal solution or the farther from the negative ideal solution, the better the solution is. The Euclidean distance between the feasible solution and the positive and negative ideal solutions is calculated as follows:

$$L^+ = \sqrt{\sum_{i=1}^{4} \omega_i \cdot (Z_i' - 1)^2} \quad (33)$$

$$L^- = \sqrt{\sum_{i=1}^{4} \omega_i \cdot (Z_i' - 0)^2} \quad (34)$$

In (33) and (34), ω_i is the weight of the ith objective, which reflects the degree of importance of the ith objective in this multi-objective optimization process. To obtain different combinations of weight coefficients, the binomial distribution of discrete probability distribution in probability theory and statistics is used in this reference paper to set the weights for the multi-objective optimization model containing n objectives. Within probability theory and statistics, the binomial distribution with parameters n and p represents the probability distribution of the number of successes of n independent trials. In each independent trial only two values are taken, and the probability of a successful value is p. Then the probability of an unsuccessful trial is 1 - p. The probability formula is expressed as follows:

$$P(X = k) = C_n^k \cdot p^k \cdot (1-p)^{n-k} \quad (35)$$

From the above equation (35), we can calculate

$$\sum_{k=0}^{n} P(X = k) = \sum_{k=0}^{n} C_n^k \cdot p^k \cdot (1-p)^{n-k} = (1 - p + p)^n = 1 \quad (36)$$

Therefore, the formula for determining the weight coefficients by binomial distribution can be expressed by equation (37).

$$(\omega_1, \omega_1, \cdots, \omega_n) = \left(C_{n-1}^0 \cdot p^0 \cdot (1-p)^{n-1}, C_{n-1}^1 \cdot p \cdot (1-p)^{n-2}, \cdots, C_{n-1}^{n-1} \cdot p^{n-1} \cdot (1-p)\right) \quad (37)$$

The above equation shows that the combination of weights obtained by the above method satisfies the sum of the weights of all objectives to be 1. By taking different values for the probability p, different combinations of weight coefficients can be obtained, thus reflecting the degree of preference of different participants in multimodal transportation for multiple objectives. If we need to set x different sets of weight combinations, the probability p=j/(x+1) for each set of weights, where j=1,2,3,...,x.

6) Converting multiple objectives into a single objective

First, the closeness of the feasible solution is calculated, and the closeness of the feasible solution can be calculated by the following equation：

$$M = \frac{L^-}{L^+ + L^-} \quad (38)$$

Obviously, the value of M ranges from (0,1), and the larger its value, the closer the current feasible solution is to the positive ideal solution and the farther it is to the negative ideal solution. Therefore, the multi-objective model of this paper can be converted into a single-objective solution, i.e., the maximum value of the closeness.

And because the closeness function satisfies the non-negative maximization characteristics of the fitness function and can reflect the degree of superiority of the corresponding feasible solution, this paper directly selects the closeness of the feasible solution to the ideal solution as the fitness function, i.e., F= M.

At this point, in summary, the design of the fitness function required to solve the multi-objective optimization model in this paper is complete.

C. Crossover and Mutation

Two parent individuals are selected according to the crossover probability Pc (generally taken as 0.6-0.9), and new individuals are derived from them by exchanging some dissimilar genes on chromosomes. The crossover probability Pc has a large impact on the performance of the genetic algorithm and directly affects the convergence of the algorithm. Although the population is more likely to have new individuals when Pc is relatively large, the retention rate of good individuals in the population decreases accordingly when it becomes large. Referring to the method in the literature [12], an adaptive mechanism is considered in this paper to adjust the crossover probability, and the adaptive crossover probability Pc is calculated as follows:

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$$Pc=\begin{cases} Pc_{\max}-(Pc_{\max}-Pc_{\min})\left(\dfrac{g}{2G}+\dfrac{f_i'-f_{avg}}{2(f_{\max}-f_{avg})}\right), f'\geq f_{avg} \\ Pc_{\max}, f'<f_{avg} \end{cases} \quad (39)$$

In equation (39):Pcmax and Pcmin are the maximum and minimum values of the crossover probability, respectively;*g* is the number of generations of the current iteration;G is the maximum number of iterative generations in the algorithm solution process;*fi* is the larger fitness value of the two bodies to be crossed;*favg* is the average fitness value of the population;*fmax* is the maximum fitness value of all individuals.

In this paper, the mutation operation is also adjusted using an adaptive mechanism. As above, the adaptive mutation probability Pm is calculated as follows:

$$Pm=\begin{cases} Pm_{\max}-(Pm_{\max}-Pm_{\min})\left(\dfrac{g}{2G}+\dfrac{f_i-f_{avg}}{2(f_{\max}-f_{avg})}\right), f\geq f_{avg} \\ Pm_{\max}, f<f_{avg} \end{cases} \quad (40)$$

In equation (40):Pmmax and Pmmin are the maximum and minimum values of the probability of mutation, respectively;*fi* is the fitness value of the individual to be mutated.

D. Termination condition

Specifies that the algorithm terminates when it reaches the maximum number of iteration generations and outputs the result.

IV. EXAMPLE ANALYSIS

A. Example Description

In this paper, we choose Chengdu as the starting point of multimodal transportation and Shanghai as the end point of multimodal transportation. Now assume that an enterprise has a batch of goods that need to be transported from Chengdu to Shanghai, starting from the zero point on the first day, with a given receiving time window of [53,77] and the number of goods set to five 20-foot full container cargoes, each container containing 20t of goods. Eleven cities upstream and downstream of the Yangtze River basin are set as transit nodes, including Luzhou, Chongqing, Yichang, Jingzhou, Wuhan, Jiujiang, Anqing, Tongling, Wuhu, Nanjing, and Nantong. In total, there are 13 node cities. The specific multimodal transportation network is shown in Fig. 3 below, in which N1-N11 represent the above 11 transit node cities in order.

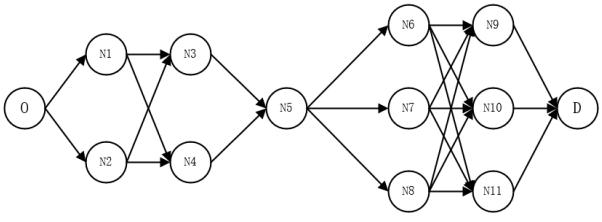


Fig. 3. Chengdu-Shanghai multimodal transport network diagram

In addition, for the convenience of presentation in programming with MATLAB software, the above 13 city nodes are replaced with Arabic numbers for numbering representation, as shown in Table 1 below.

TABLE I. CITY NUMBERING CROSS-REFERENCE TABLE

Number	Representative Cities	Number	Representative Cities
1	Chengdu	9	Tongling
2	Luzhou	10	Wuhu
3	Chongqing	11	Nanjing
4	Yichang	12	Nantong
5	Jingzhou	13	Shanghai
6	Wuhan		
7	Jiujiang		
8	Anqing		

The speed of three modes of transportation — road, railroad, and water—is adopted as the empirical average. In this paper, the average speed of road transportation is set at 65 km per hour, the average speed of railroad transportation is 75 km per hour, and the average speed of water transportation is 30 km per hour. The transportation distance, transportation time, and transportation cost between different cities are shown in Tables 2, 3, and 4, respectively.

TABLE II. TABLE OF TRANSPORTATION DISTANCE BETWEEN DIFFERENT CITIES

Route starting point	Route end point	Transportation distance/km		
		Road Transportation	*Rail Transportation*	*Waterway transportation*
1	2	261	281	/
1	3	313	304	/
2	4	737	/	782
2	5	831	/	908
3	4	573	553	570
3	5	667	641	693
4	6	320	292	405
5	6	221	204	266
6	7	239	252	214
6	8	347	335	356
6	9	455	431	452
7	10	366	325	357
7	11	465	450	455
7	12	694	910	701
8	10	192	172	190
8	11	286	257	295
8	12	476	/	537
9	10	94	78	108
9	11	187	203	198
9	12	377	487	433
10	13	357	426	446
11	13	306	315	346
12	13	128	166	117

TABLE III. TRANSPORTATION SCHEDULE BETWEEN DIFFERENT CITIES

Route starting point	Route end point	Transportation time/h		
		Road Transportation	*Rail Transportation*	*Waterway transportation*
1	2	4.02	3.75	/
1	3	4.82	4.05	/
2	4	11.34	/	26.07
2	5	12.78	/	30.27
3	4	8.82	7.37	19.00
3	5	10.26	8.55	23.10
4	6	4.92	3.89	13.50
5	6	3.40	2.72	8.87
6	7	3.68	3.36	7.13
6	8	5.34	4.47	11.87
6	9	7.00	5.75	15.07
7	10	5.63	4.33	11.90
7	11	7.15	6.00	15.17

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7	12	10.68	12.13	23.37
8	10	2.95	2.29	6.33
8	11	4.40	3.43	9.83
8	12	7.32	/	17.90
9	10	1.45	1.04	3.60
9	11	2.88	2.71	6.60
9	12	5.80	6.49	14.43
10	13	5.49	5.68	14.87
11	13	4.71	4.20	11.53
12	13	1.97	2.21	3.90

TABLE IV. TABLE OF UNIT TRANSPORTATION COSTS BETWEEN DIFFERENT CITIES

Route starting point	Route end point	Unit transportation cost/（yuan/TEU）		
		Road Transportation	*Rail Transportation*	*Waterway transportation*
1	2	1596.00	1334.99	/
1	3	1908.00	1408.24	/
2	4	4452.00	/	1173.00
2	5	5016.00	/	1362.00
3	4	3468.00	2201.31	855.00
3	5	4032.00	2481.59	1039.50
4	6	1950.00	1370.02	607.50
5	6	1356.00	1089.74	399.00
6	7	1464.00	1242.62	321.00
6	8	2112.00	1506.98	534.00
6	9	2760.00	1812.74	678.00
7	10	2226.00	1475.13	535.50
7	11	2820.00	1873.25	682.50
7	12	4194.00	3338.35	1051.50
8	10	1182.00	987.82	285.00
8	11	1746.00	1258.55	442.50
8	12	2886.00	/	805.50
9	10	594.00	688.43	162.00
9	11	1152.00	1086.56	297.00
9	12	2292.00	1991.10	649.50
10	13	2172.00	1796.81	669.00
11	13	1866.00	1443.28	519.00
12	13	798.00	968.71	175.50

The transit times as well as transit costs are shown in Tables 5 and 6.

TABLE V. CHANGEOVER TIME BETWEEN DIFFERENT TRANSPORTATION MODES IN EACH NODE CITY (H/TEU)

Mode of Transportation	Road	Rail	Waterway
Road	0	0.35	0.4
Rail	0.35	0	0.5
Waterway	0.4	0.5	0

TABLE VI. TRANSIT COSTS BETWEEN DIFFERENT MODES OF TRANSPORT IN EACH NODE CITY (YUAN/TEU)

Mode of Transportation	Road	Rail	Waterway
Road	0	100	200
Rail	100	0	250
Waterway	200	250	0

The data related to carbon emissions are shown in Tables 7 and 8.

TABLE VII. CO_2 EMISSIONS PER UNIT FOR DIFFERENT MODES OF TRANSPORT

Mode of Transportation	Unit CO2 emissions（kg-CO2/TEU·km）
Road Transportation	0.9214
Rail Transportation	0.1838
Waterway transportation	0.0941

TABLE VIII. TABLE OF CARBON EMISSIONS PER UNIT OF CONVERSION BETWEEN DIFFERENT TRANSPORTATION MODES

Mode of Transportation	Unit CO2 emissions（kg-CO2/TEU）
Road Transportation	2.56
Rail Transportation	2.34
Waterway transportation	2.26

B. Solution and Analysis of Results

In this paper, MATLAB 2019A software is used to program the model for solution, and after adjusting the parameters, the population size of the improved genetic algorithm is determined as popsize = 20, the maximum number of iterations maxgen = 100, the maximum crossover probability Pcmax = 0.9, the minimum crossover probability Pcmin = 0.6, the maximum variation probability Pmmax = 0.1, the minimum variation probability Pmmin = 0.001, a = 70, and b = 100 in the penalty cost function. Assume that the departure time of railroad transportation starts at 6:00 a.m. and ends at 10:00 p.m. every day with an 8-hour departure period, and the departure time of waterway transportation starts at 7:00 a.m. and ends at 11:00 p.m. every day with an 8-hour departure period. The waiting cost of cargo containers in the node city is set at 50 yuan per h-TEU. The average cargo loss rate is 0.01% per 100 km when cargo is transported by road, 0.015% per 100 km when transported by railroad, 0.02% per 100 km when transported by waterway, and 0.04% per time when a transshipment operation is carried out.

The binomial distribution assignment method described above is used to set the weights for the four objectives, and this paper is set to obtain 15 sets of weight combinations, with the probability p = (1/16, 2/16,..., 14/16, 15/16) corresponding to each set of weight combinations. The model under different weight combinations is solved using Matlab 2019A, and the optimal multimodal transportation scheme under the corresponding weight combinations is derived, as shown in Table 9 below.

TABLE IX. OPTIMAL TRANSPORTATION SCHEME UNDER DIFFERENT WEIGHT COMBINATIONS

Probabilityp	Weighting combinations	Proximity value	Optimal transportation solution
1/16	（0.82397,0.16479,0.01099,0.00024）	0.8001	1 —road→ 3 —waterway→ 5 —road→ 6 —waterway→ 7 —waterway→ 11 —waterway→ 13
2/16	（0.66992,0.28711,0.04102,0.00195）	0.7227	1 —rail→ 3 —rail→ 4 —rail→ 6 —waterway→ 7 —waterway→ 11 —waterway→ 13
3/16	（0.53638,0.37134,0.08569,0.00659）	0.6967	1 —rail→ 3 —rail→ 4 —rail→ 6 —waterway→ 7 —waterway→ 11 —waterway→ 13
4/16	（0.42188,0.42188,0.14063,0.01563）	0.6787	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 9 —waterway→ 11 —waterway→ 13

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5/16	（0.3249 5,0.4431 2,0.2014 2,0.0305 2）	0.7394	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 9 —waterway→ 11 —waterway→ 13
6/16	（0.2441 4,0.4394 5,0.2636 7,0.0527 3）	0.6736	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 9 —waterway→ 11 —waterway→ 13
7/16	（0.1779 8,0.4152 8,0.3230 0,0.0837 4）	0.7020	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 9 —waterway→ 11 —waterway→ 13
8/16	（0.1250 0,0.3750 0,0.3750 0,0.1250 0）	0.7306	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 11 —rail→ 13
9/16	（0.0837 4,0.3230 0,0.4152 8,0.1779 8）	0.7594	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 11 —rail→ 13
10/16	（0.0527 3,0.2636 7,0.4394 5,0.2441 4）	0.7887	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 11 —rail→ 13
11/16	（0.0305 2,0.2014 2,0.4431 2,0.3249 5）	0.8595	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 10 —rail→ 13
12/16	（0.0156 3,0.1406 3,0.4218 8,0.4218 8）	0.8491	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 10 —rail→ 13
13/16	（0.0065 9,0.0856 9,0.3713 4,0.5363 8）	0.8808	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 10 —rail→ 13
14/16	（0.0019 5,0.0410 2,0.2871 1,0.6699 2)	0.9139	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 10 —rail→ 13
15/16	（0.0002 4,0.0109 9,0.1647 9,0.8239 7）	0.9564	1 —rail→ 3 —rail→ 4 —rail→ 6 —rail→ 8 —rail→ 11 —rail→ 13

From the results in Table 9 above, we can see that when p = 1/16, the weight of transportation cost among the four objectives is taken as the maximum value. At this time, the transportation scheme is road-water transportation; the goods depart from the origin node 1 by road transportation to node 3, then switch to water transportation to node 5, then by road transportation to node 6, and finally switch to water transportation at node 6 all the way to the end point 13. At this time, the reason why the scheme was not chosen in transit The reason for not choosing the railroad and waterway transportation with lower transportation costs but choosing the road transportation is that the schedule constraint means that the time spent waiting for the schedule is too long, which leads to excessive waiting costs. The choice of railroad transportation will have a negative effect and increase the overall transportation cost, so the iron-water transportation scheme is not chosen. Therefore, it is important to take the schedule factor into account in the process of finding the optimal multimodal transport solution.

When the value of p increases from 1/16 to 2/16, the mode of transportation chosen for the optimal transport solution changes to rail-water intermodal transport, where the goods are transported from the origin node 1 to node 6 by rail and then converted to waterway transport to the end point 13.

When the value of p is gradually increased from 2/16 to 8/16, the weight of the transportation cost target gradually decreases, the weight of the transportation time and carbon emission target gradually increases, and the proportion of railroad transportation produces an increase, which is caused by the shorter time consumption of railroad transportation than water transportation, but the overall transportation scheme is still combined railroad and water transportation.

When the value of p is 8/16, the optimal transportation scheme is changed from rail-water transportation to full railroad transportation, and the goods are transported from the origin node 1 to the end point 13 by railroad. When the value of p is increased from 9/16 to 11/16, the weights of transportation time and carbon emission objectives are still higher, and the optimal transportation scheme remains unchanged at this time and is still full railroad transportation. In the case of considering transportation time and carbon emissions, the advantage of railroad transportation is much greater than the remaining two. When the value of p increases from 10/16 to 15/16, the weight of logistics service quality objectives gradually increases and reaches its maximum at p = 15/16, at which time the optimal transportation solution is still full railroad transportation because the logistics service quality in this paper is measured by timeliness and safety, and railroad transportation has a greater advantage in the logistics service quality indexes defined in this paper with its faster speed and lower cargo damage rate.

In addition, although the speed of road transport is faster and the rate of cargo damage is lower, because general multimodal shippers have requirements for the receipt time, the goods need to arrive at the destination within a given time window; therefore, under the set time window, road transport arriving at the destination too early will reduce customer satisfaction in terms of time, which will reduce the customer's perceived level of logistics services. If the time window is advanced or the customer's demand for safety is increased, i.e., a lower rate of cargo damage is required, then road transport has an advantage over rail transport in terms of logistics service quality.

Moreover, the weight combinations listed above do not represent the different needs of all decision-makers in the

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multimodal transport problem for each objective. For example, due to the nature of the binomial coefficients, the weight combinations set above do not have weight combinations that focus on both transport costs and carbon emissions, so the weight combination W=(0.4,0.1,0.4,0.1) is set, and under such weight combinations, the optimal solution is solved by using The optimal transportation scheme is $1 \xrightarrow{rail} 3 \xrightarrow{rail} 4 \xrightarrow{rail} 6 \xrightarrow{waterway} 9 \xrightarrow{waterway} 11 \xrightarrow{waterway} 13$. That is, the goods from the origin node 1, through the railroad transport to node 6, through the intermediate conversion operations, will be transported to the waterway transport at the end of 13. This fully reflects the important role of rail-water transport in reducing transport costs and carbon emissions.

V. Conclusion

Based on the analysis of the objective function, this paper constructs a multi-objective path optimization model for multimodal container transportation with schedule constraints, considering the four factors of transportation time, transportation cost, carbon emission, and multimodal logistics service quality, and designs an improved genetic algorithm to transform the multi-objective optimization problem into a single-objective optimization problem with closeness as the objective. The results show that the model and algorithm constructed in this paper can better balance the relationship between multiple objectives and can effectively provide a high-practical optimization scheme and route reference for container multimodal transport route optimization. However, in order to simplify the calculation, this paper assumes many virtual conditions when constructing the route optimization model, such as not considering the transportation capacity of the transportation mode, but in the actual situation, if too much cargo needs to be transported, there may be a problem that the road and railroad transportation capacity is not enough to complete the transportation at one time. Therefore, the model and the algorithm designed in this paper have some limitations in the real world, and the model algorithm can be better optimized and improved according to the actual situation in the future.

Acknowledgment

Throughout the writing of this dissertation, I have received a great deal of support and assistance. I would like to acknowledge Professor Chenning for her valuable guidance throughout my studies.

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